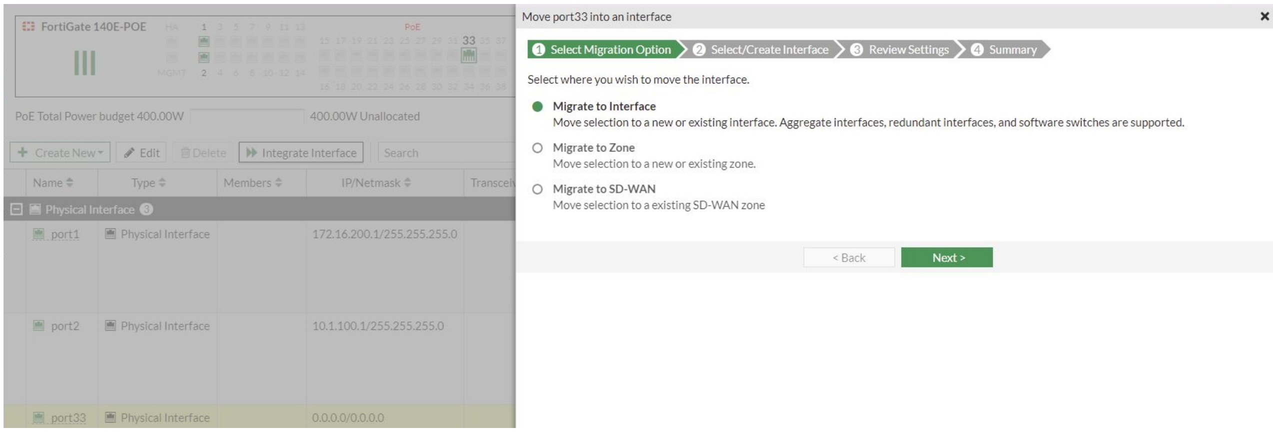
#### To integrate an interface:

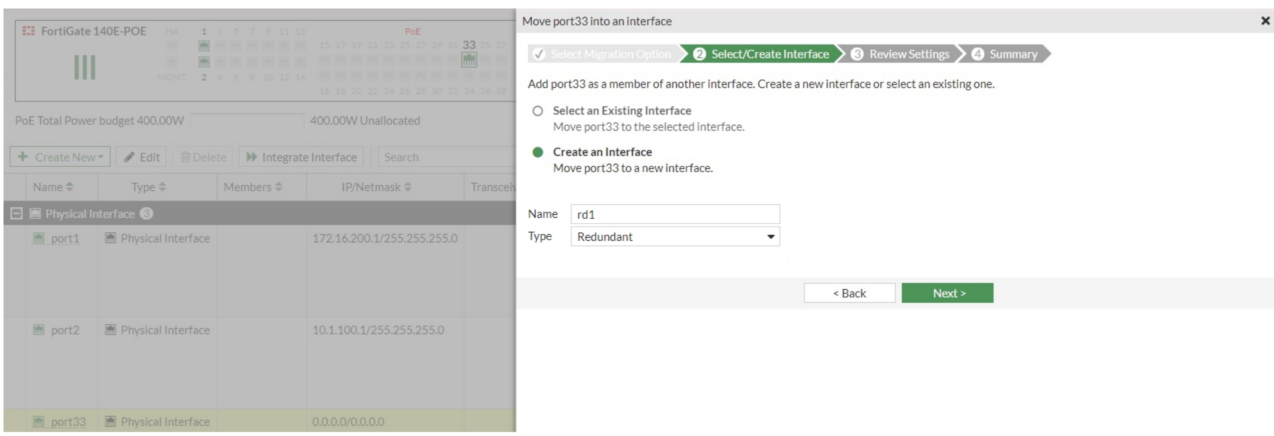
1. Go to *Network* > *Interfaces* and select an interface in the list.
2. Click *Integrate Interface*. The wizard opens.

   > Alternatively, select an interface in the list. Then right-click and select *Integrate Interface*.

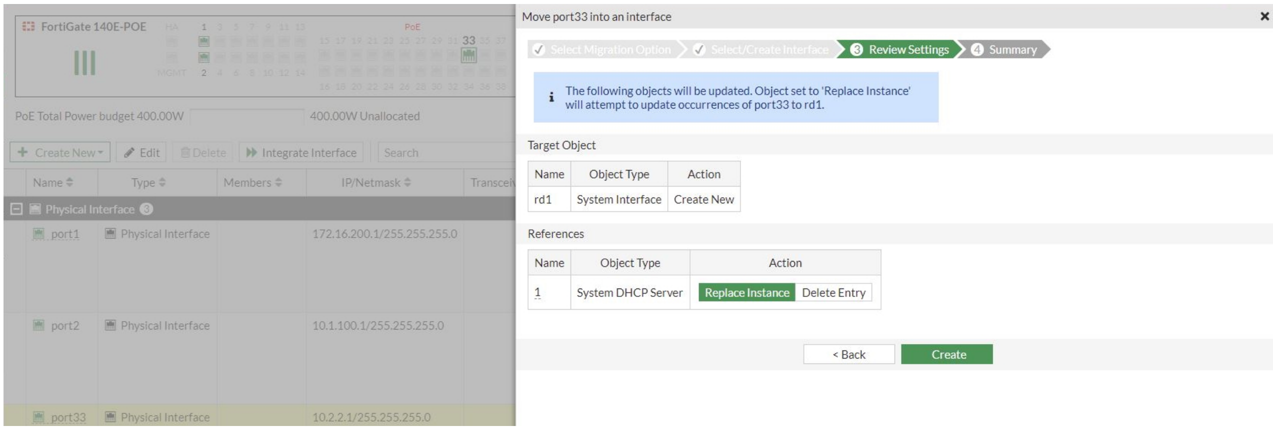
3. Select *Migrate to Interface* and click *Next*.



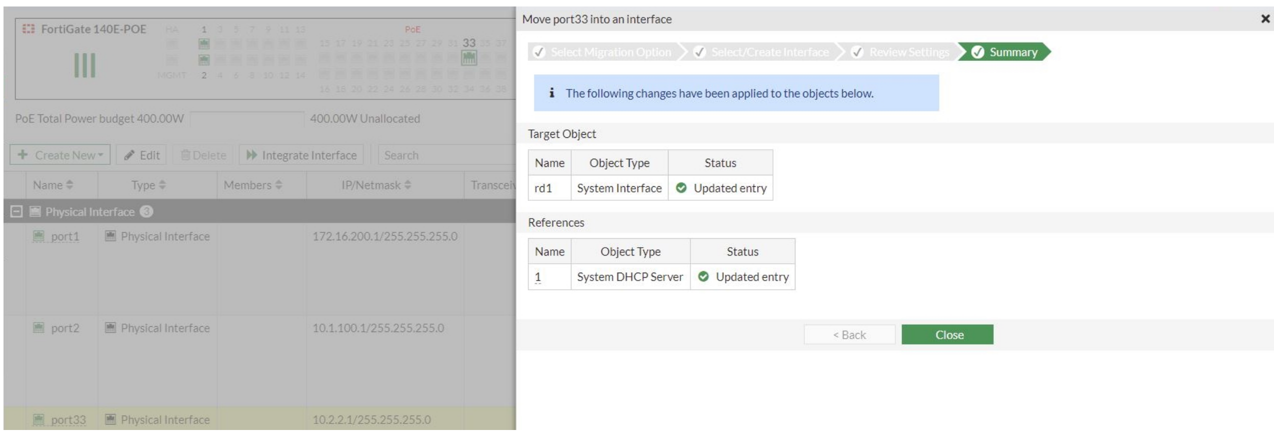

4. Select *Create an Interface*. Enter a name (*rd1*) and set the *Type* to *Redundant*.




5. Click *Next*. The *References* sections lists the associated services with options to *Replace Instance* or *Delete Entry*.
6. For the DHCP server *Action*, select *Replace Instance* and click *Create*.




7. The migration occurs automatically and the statuses for the object and reference change to *Updated entry*. Click *Close*.

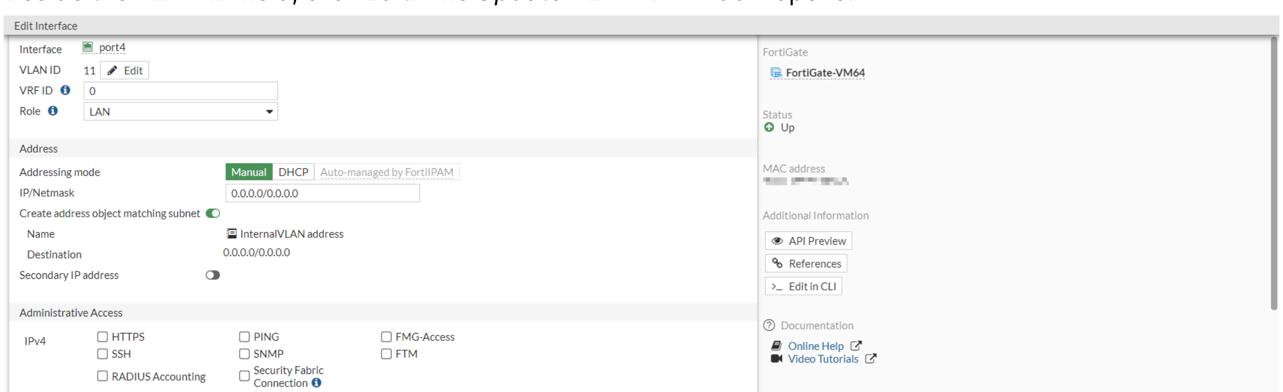



## Changing the VLAN ID

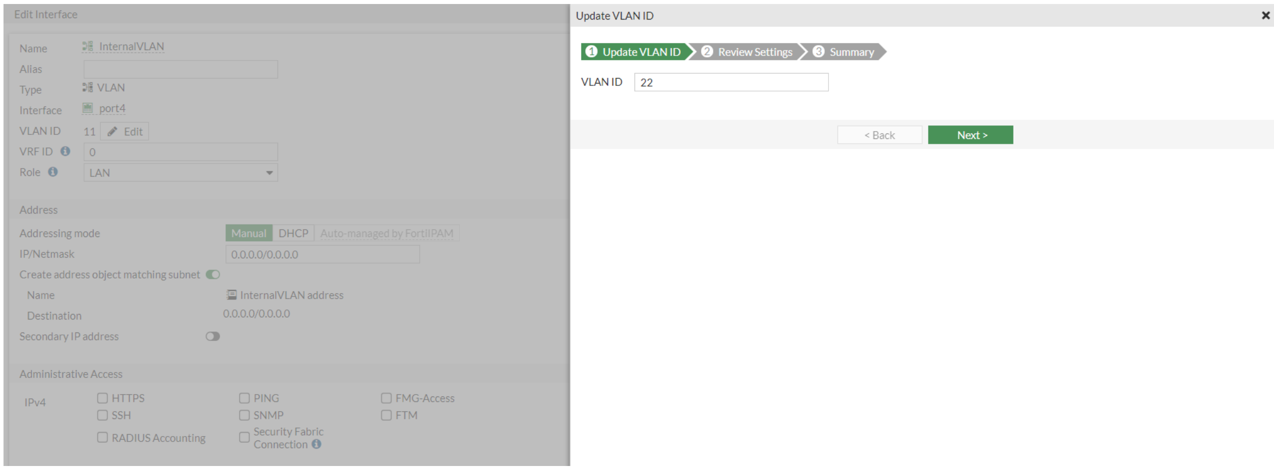
In this example, the VLAN ID of *InternalVLAN* is changed from 11 to 22.

### To change the VLAN ID:

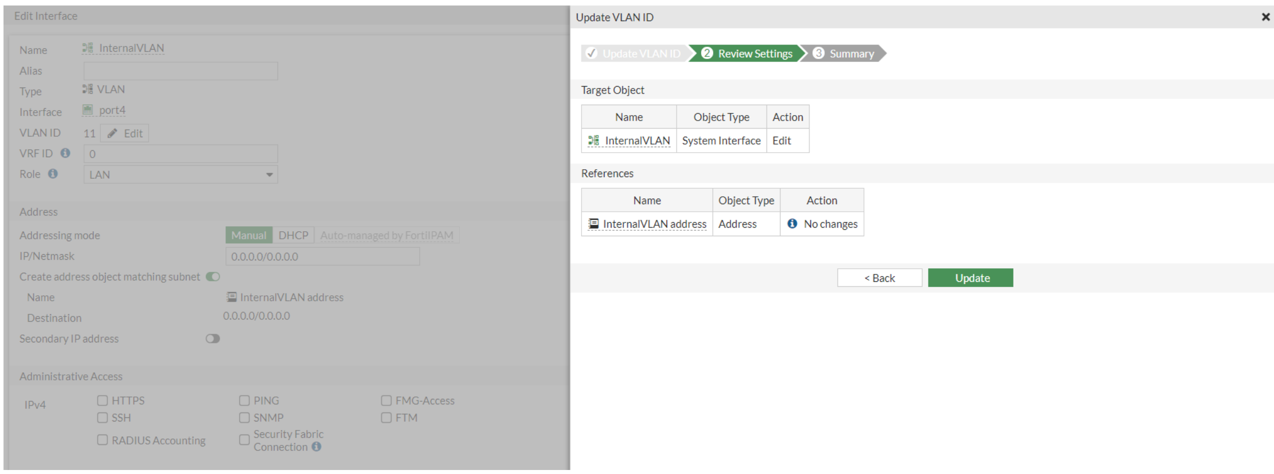
1. Go to *Network > Interfaces* and edit an existing interface.
2. Beside the *VLAN ID* field, click *Edit*. The *Update VLAN ID* window opens.



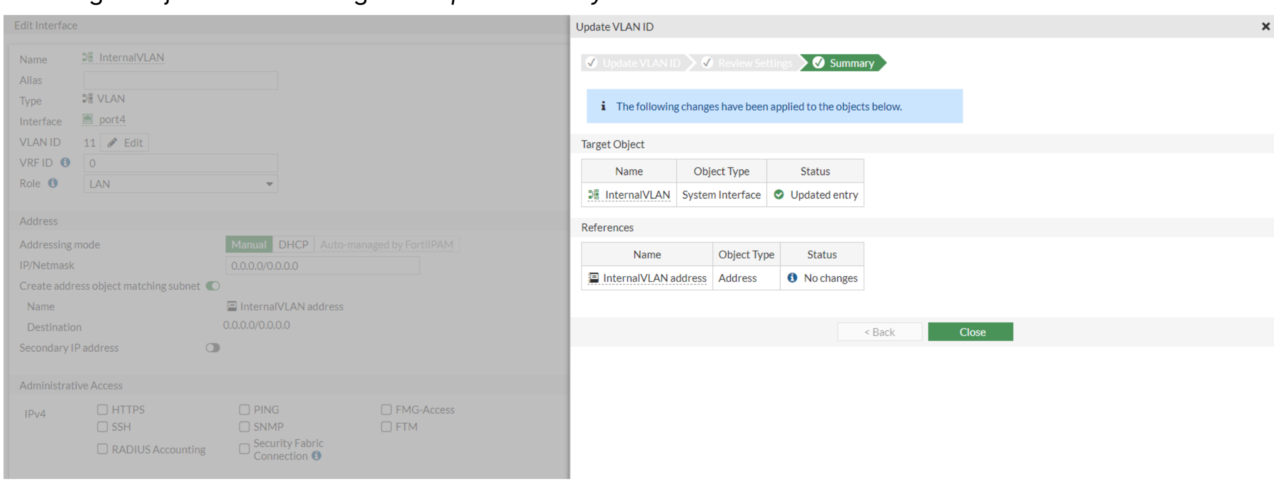

3. Enter the new ID (*22*) and click *Next*.



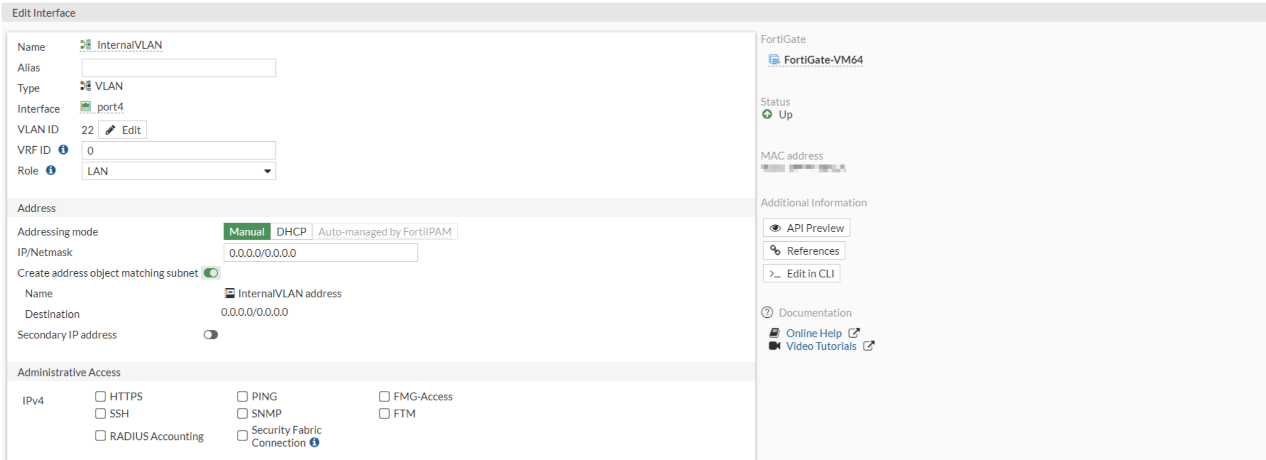

4. Verify the changes, then click *Update* and *OK*.




5. The target object status changes to *Updated entry*. Click *Close*.




In the interface settings, the ID displays as *22*.




# Captive portals

A captive portal is used to enforce authentication before web resources can be accessed. Until a user authenticates successfully, any HTTP request returns the authentication page. After successfully authenticating, a user can access the requested URL and other web resources, as permitted by policies. The captive portal can also be configured to only allow access to members of specific user groups.

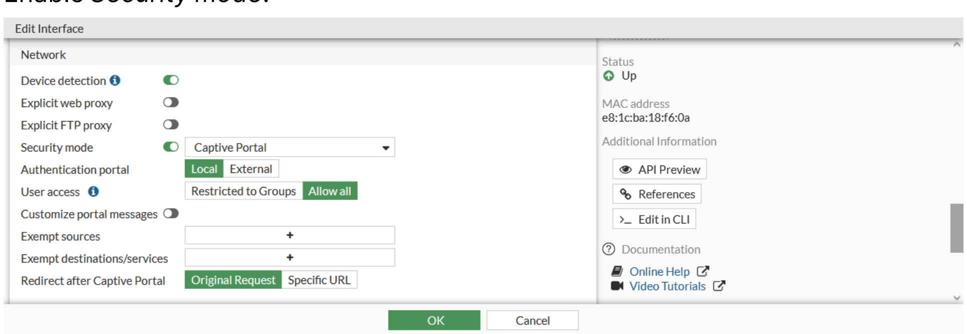
Captive portals can be hosted on the FortiGate or an external authentication server. They can be configured on any network interface, including VLAN and WiFi interfaces. On a WiFi interface, the access point appears open, and the client can connect to access point with no security credentials, but then sees the captive portal authentication page. See Captive Portal Security, in the FortiWiFi and FortiAP Configuration Guide for more information.

All users on the interface are required to authenticate. Exemption lists can be created for devices that are unable to authenticate, such as a printer that requires access to the internet for firmware upgrades.

> When configuring a bridge mode SSID, you do not need to enable captive portal.

**To configure a captive portal in the GUI:**

1. Go to *Network > Interfaces* and edit the interface that the users connect to. The interface *Role* must be *LAN* or *Undefined*.
2. Enable *Security mode*.




3. Configure the following settings, then click *OK*.

| | |
|---|---|
| **Authentication Portal** | Configure the location of the portal:<br>• *Local*: the portal is hosted on the FortiGate unit.<br>• *External*: enter the FQDN or IP address of external portal. |
| **User access** | Select if the portal applies to all users, or selected user groups:<br>• *Restricted to Groups*: restrict access to the selected user groups. The *Login page* is shown when a user tries to log in to the captive portal.<br>• *Allow all*: all users can log in, but access will be defined by relevant policies. The *Disclaimer page* is shown when a user tried to log in to the captive portal. |
| **Customize portal messages** | Enable to use custom portal pages, then select a replacement message group. See Custom captive portal pages on page 249. |
| **Exempt sources** | Select sources that are exempt from the captive portal.<br>Each exemption is added as a rule in an automatically generated exemption list. |
| **Exempt destinations/services** | Select destinations and services that are exempt from the captive portal.<br>Each exemption is added as a rule in an automatically generated exemption list. |
| **Redirect after Captive Portal** | Configure website redirection after successful captive portal authentication:<br>• *Original Request*: redirect to the initially browsed to URL .<br>• *Specific URL*: redirect to the specified URL.<br>The *Redirect after Captive Portal* settings are ignored when the `CONTINUE_URL = "<web site>"` parameter is included in the captive portal request. |

**To configure a captive portal in the CLI:**

1. If required, create a security exemption list:

```
config user security-exempt-list
    edit <list>
        config rule
            edit 1
                set srcaddr <source(s)>
                set dstaddr <source(s)>
                set service <service(s)>
            next
            edit 2
                set srcaddr <source(s)>
                set dstaddr <source(s)>
                set service <service(s)>
            next
        end
    next
end
```

2. Configure captive portal authentication on the interface:

```
config system interface
    edit <interface>
        set security-mode {none | captive-portal}
        set security-external-web <string>
        set replacemsg-override-group <group>
        set security-redirect-url <string>
        set security-exempt-list <list>
        set security-groups <group(s)>
    next
end
```

### Custom captive portal pages

Portal pages are HTML files that can be customized to meet user requirements.

Most of the text and some of the HTML in the message can be changed. Tags are enclosed by double percent signs (%%); most of them should not be changed because they might carry information that the FortiGate unit needs. For information about customizing replacement messages, see Modifying replacement messages on page 4087.

The images on the pages can be replaced. For example, your organization's logo can replace the Fortinet logo. For information about uploading and using new images in replacement messages, see Replacement message images on page 4088.

The following pages are used by captive portals:

| | |
|---|---|
| **Login Page** | Requests user credentials.<br>The *%%QUESTION%%* tag provides the *Please enter the required information to continue.* text.<br>This page is shown to users that are trying to log in when *User access* is set to *Restricted to Groups*. |
| **Login Failed Page** | Reports that incorrect credentials were entered, and requests correct credentials.<br>The *%%FAILED_MESSAGE%%* tag provides the *Firewall authentication failed. Please try again.* text. |
| **Disclaimer Page** | A statement of the legal responsibilities of the user and the host organization that the user must agree to before proceeding. This page is shown users that are trying to log in when *User access* is set to *Allow all*. |
| **Declined Disclaimer Page** | Shown if the user does not agree to the statement on the Disclaimer page. Access is denied until the user agrees to the disclaimer. |

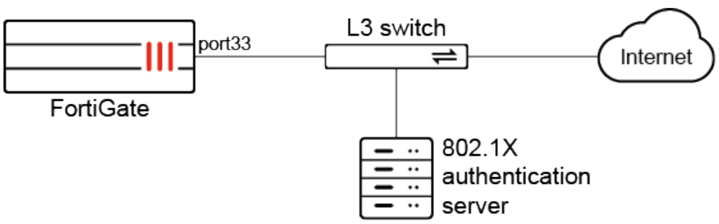
## Configuring a FortiGate interface to act as an 802.1X supplicant

A FortiGate interface can be configured to act as a 802.1X supplicant. The settings can be enabled on the network interface in the CLI. The EAP authentication method can be either PEAP or TLS using a user certificate.

```
config system interface
    edit <interface>
        set eap-supplicant {enable | disable}
        set eap-method {peap | tls}
        set eap-identity <identity>
        set eap-password <password>
        set eap-ca-cert <CA_cert>
        set eap-user-cert <user_cert>
    next
end
```

## Example

In this example, the FortiGate connects to an L3 switch that is not physically secured. All devices that connect to the internet through the L3 switch must be authenticated with 802.1X on the switch port by either a username and password (PEAP), or a user certificate (TLS). Configuration examples for both EAP authentication methods on port33 are shown.




**To configure EAP authentication with PEAP:**

1. Configure the interface:

```
config system interface
    edit "port33"
        set vdom "vdom1"
        set ip 7.7.7.2 255.255.255.0
        set allowaccess ping https ssh snmp http telnet fgfm radius-acct probe-response fabric
        set stpforward enable
        set type physical
        set snmp-index 42
        set eap-supplicant enable
        set eap-method peap
        set eap-identity "test1"
        set eap-password **********
    next
end
```

2. Verify the interface's PEAP authentication details:

```
# diagnose test app eap_supp 2
Interface: port33
status:Authorized
method: PEAP
identity: test1
```

```
ca_cert:
client_cert:
private_key:
last_eapol_src =70:4c:a5:3b:0b:c6
```

Traffic is able to pass because the status is authorized.

**To configure EAP authentication with TLS:**

1. Configure the interface:

```
config system interface
    edit "port33"
        set vdom "vdom1"
        set ip 7.7.7.2 255.255.255.0
        set allowaccess ping https ssh snmp http telnet fgfm radius-acct probe-response fabric
        set stpforward enable
        set type physical
        set snmp-index 42
        set eap-supplicant enable
        set eap-method tls
        set eap-identity "test2@fortiqa.net"
        set eap-ca-cert "root_G_CA_Cert_1.cer"
        set eap-user-cert "root_eap_client_global.cer"
    next
end
```

2. Verify the interface's TLS authentication details:

```
# diagnose test application eap_supp 2
Interface: port33
status:Authorized
method: TLS
identity: test2@fortiqa.net
ca_cert: /etc/cert/ca/root_G_CA_Cert_1.cer
client_cert: /etc/cert/local/root_eap_client_global.cer
private_key: /etc/cert/local/root_eap_client_global.key
last_eapol_src =70:4c:a5:3b:0b:c6
```

Traffic is able to pass because the status is authorized.

## Auto speed negotiation for 10G Base-T on FortiGate 100xF devices

Auto speed negotiation on the 10G Base-T interface allows the 1G/10G copper ports on the FortiGate 100xF devices to automatically handle both 1G and 10G speeds and duplex settings, eliminating the need for manual adjustments.

```
config system interface
    edit "port1"
        set speed auto
```

```
    next
end
```

## Example

- When the peer uses 1G and the speed is the default value:

```
config system interface
    edit "port1"
        set vdom "root"
        set type physical
        set speed 10000auto
    next
end
```

The port is down because of the different speeds:

```
# get sys int physical port1
== [onboard]
        ==[port1]
                mode: static
                ip: 0.0.0.0 0.0.0.0
                ipv6: ::/0
                status: down
                speed: n/a
                FEC: none
                FEC_cap: {none}
```

- After the speed is set to auto:

```
config system interface
    edit "port1"
        set speed auto
    next
end
```

The port is up and auto negotiates to the correct speed and duplex setting:

```
# get sys int physical port1
== [onboard]
        ==[port1]
                mode: static
                ip: 0.0.0.0 0.0.0.0
                ipv6: ::/0
                status: up
                speed: 1000Mbps (Duplex: full)
                FEC: none
                FEC_cap: {none}
```

# Customizable CoS marking for locally generated ARP packets (NEW)

FortiGate supports 802.1p class of service (CoS) markings (cos0-cos7) for locally generated ARP packets, allowing customers to align ARP traffic with network QoS policies and constraints.

**To set the CoS in VLAN tag for outgoing ARP packets:**

```
config system interface
     edit <interface>
        set arp-egress-cos {cos0 | cos1 | cos2 | cos3 | cos4 | cos5 | cos6 | cos7}
    next
end
```

## Example


A FortiGate and PC are both configured to be in VLAN66. The FortiGate VLAN interface is configured to send ARP packets with a CoS marking of 7, denoting the Network Control priority level.

**To test the CoS markings in locally generated ARP packets:**

1. Configure the interface on the FortiGate:

```
config system interface
     edit "v6"
        set vdom "root"
        set ip 16.1.1.1 255.255.255.0
        set allowaccess ping
        set arp-egress-cos cos7
        set snmp-index 40
        set ip-managed-by-fortiipam disable
        set interface "port6"
        set vlanid 66
    next
end
```

2. Configure the PC network:

```
~$ sudo cat /etc/netplan/01-network-manager-all.yaml
...
  vlans:
    vlan66:
      id: 66
      link: ens160
      addresses:
        - 16.1.1.66/24
```

3. Ping the FortiGate from the PC:

```
fosqa@fosqa-Cos:~$ ping 16.1.1.1 -c 2
PING 16.1.1.1 (16.1.1.1) 56(84) bytes of data.
64 bytes from 16.1.1.1: icmp_seq=1 ttl=255 time=0.193 ms
64 bytes from 16.1.1.1: icmp_seq=2 ttl=255 time=0.117 ms

--- 16.1.1.1 ping statistics ---
2 packets transmitted, 2 received, 0% packet loss, time 1038ms
rtt min/avg/max/mdev = 0.117/0.155/0.193/0.038 ms
```

4. Do a packet capture on the PC:
   An ARP request sent from the FortiGate sets its priority to `Network Control (7)` inside the VLAN header:

```
fosqa@fosqa-Cos:~$ sudo tshark -i ens160 -Y "vlan" -V
Running as user "root" and group "root". This could be dangerous.
Capturing on 'ens160'
...
Frame 2: 64 bytes on wire (512 bits), 64 bytes captured (512 bits) on interface ens160, id 0
    Section number: 1
    Interface id: 0 (ens160)
        Interface name: ens160
    Encapsulation type: Ethernet (1)
    Arrival Time: Sep 15, 2025 16:53:41.041847718 PDT
    UTC Arrival Time: Sep 15, 2025 23:53:41.041847718 UTC
    Epoch Arrival Time: 1757980421.041847718
    [Time shift for this packet: 0.000000000 seconds]
    [Time delta from previous captured frame: 0.000111729 seconds]
    [Time delta from previous displayed frame: 0.000111729 seconds]
    [Time since reference or first frame: 0.000111729 seconds]
    Frame Number: 2
    Frame Length: 64 bytes (512 bits)
    Capture Length: 64 bytes (512 bits)
    [Frame is marked: False]
    [Frame is ignored: False]
    [Protocols in frame: eth:ethertype:vlan:ethertype:arp]
Ethernet II, Src: Fortinet_15:08:99 (94:f3:92:15:08:99), Dst: Broadcast (ff:ff:ff:ff:ff:ff)
    Destination: Broadcast (ff:ff:ff:ff:ff:ff)
        Address: Broadcast (ff:ff:ff:ff:ff:ff)
        .... ..1. .... .... .... .... = LG bit: Locally administered address (this is NOT the
factory default)
        .... ...1 .... .... .... .... = IG bit: Group address (multicast/broadcast)
    Source: Fortinet_15:08:99 (94:f3:92:15:08:99)
        Address: Fortinet_15:08:99 (94:f3:92:15:08:99)
        .... ..0. .... .... .... .... = LG bit: Globally unique address (factory default)
        .... ...0 .... .... .... .... = IG bit: Individual address (unicast)
    Type: 802.1Q Virtual LAN (0x8100)
802.1Q Virtual LAN, PRI: 7, DEI: 0, ID: 66
    111. .... .... .... = Priority: Network Control (7)
    ...0 .... .... .... = DEI: Ineligible
    .... 0000 0100 0010 = ID: 66
    Type: ARP (0x0806)
    Padding: 0000000000000000000000000000
    Trailer: 00000000
```

```
Address Resolution Protocol (request)
    Hardware type: Ethernet (1)
    Protocol type: IPv4 (0x0800)
    Hardware size: 6
    Protocol size: 4
    Opcode: request (1)
    Sender MAC address: Fortinet_15:08:99 (94:f3:92:15:08:99)
    Sender IP address: 16.1.1.1
    Target MAC address: 00:00:00_00:00:00 (00:00:00:00:00:00)
    Target IP address: 16.1.1.66
```

An ARP reply sent by the FortiGate in response to the PC's ARP also sets its priority to `Network Control (7)`:

```
Frame 8: 64 bytes on wire (512 bits), 64 bytes captured (512 bits) on interface ens160, id 0
    Section number: 1
    Interface id: 0 (ens160)
        Interface name: ens160
    Encapsulation type: Ethernet (1)
    Arrival Time: Sep 15, 2025 16:53:46.239793531 PDT
    UTC Arrival Time: Sep 15, 2025 23:53:46.239793531 UTC
    Epoch Arrival Time: 1757980426.239793531
    [Time shift for this packet: 0.000000000 seconds]
    [Time delta from previous captured frame: 0.000082934 seconds]
    [Time delta from previous displayed frame: 0.000082934 seconds]
    [Time since reference or first frame: 5.198057542 seconds]
    Frame Number: 8
    Frame Length: 64 bytes (512 bits)
    Capture Length: 64 bytes (512 bits)
    [Frame is marked: False]
    [Frame is ignored: False]
    [Protocols in frame: eth:ethertype:vlan:ethertype:arp]
Ethernet II, Src: Fortinet_15:08:99 (94:f3:92:15:08:99), Dst: VMware_7c:6c:08
(00:0c:29:7c:6c:08)
    Destination: VMware_7c:6c:08 (00:0c:29:7c:6c:08)
        Address: VMware_7c:6c:08 (00:0c:29:7c:6c:08)
        .... ..0. .... .... .... .... = LG bit: Globally unique address (factory default)
        .... ...0 .... .... .... .... = IG bit: Individual address (unicast)
    Source: Fortinet_15:08:99 (94:f3:92:15:08:99)
        Address: Fortinet_15:08:99 (94:f3:92:15:08:99)
        .... ..0. .... .... .... .... = LG bit: Globally unique address (factory default)
        .... ...0 .... .... .... .... = IG bit: Individual address (unicast)
    Type: 802.1Q Virtual LAN (0x8100)
802.1Q Virtual LAN, PRI: 7, DEI: 0, ID: 66
    111. .... .... .... = Priority: Network Control (7)
    ...0 .... .... .... = DEI: Ineligible
    .... 0000 0100 0010 = ID: 66
    Type: ARP (0x0806)
    Padding: 0000000000000000000000000000
    Trailer: 00000000
Address Resolution Protocol (reply)
    Hardware type: Ethernet (1)
```

```
    Protocol type: IPv4 (0x0800)
    Hardware size: 6
    Protocol size: 4
    Opcode: reply (2)
    Sender MAC address: Fortinet_15:08:99 (94:f3:92:15:08:99)
    Sender IP address: 16.1.1.1
    Target MAC address: VMware_7c:6c:08 (00:0c:29:7c:6c:08)
    Target IP address: 16.1.1.66
```

# Proxy ARP on VLAN interface NEW

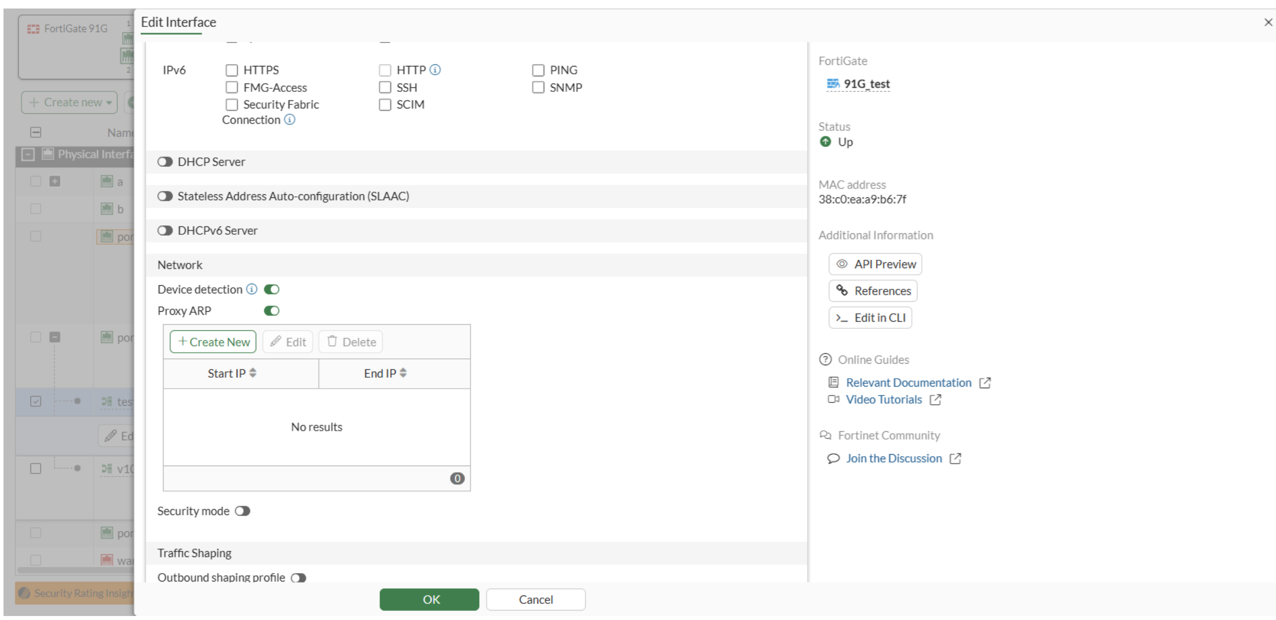
Proxy ARP address ranges for VLAN type interfaces can be added with the GUI.

In a micro-segmentation scenario, you can enable blocking intra-VLAN traffic on a FortiSwitch and utilize proxy ARP and firewall policy to define which hosts are allowed to respond to each other. See Blocking intra-VLAN traffic.

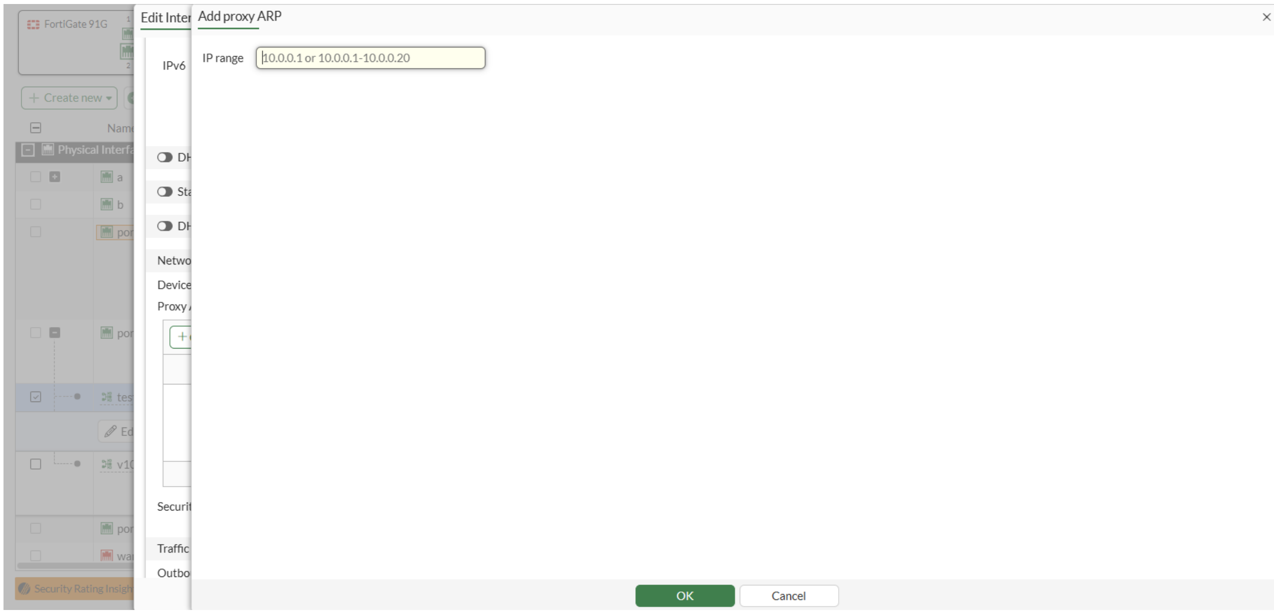
Administrators can use the proxy ARP configuration option in a VLAN interface to create, edit, and delete these address ranges.

**To configure:**

1. Go to *Network > Interfaces* and edit a VLAN interface to enable Proxy ARP.




2. Click *Create New* in the *Proxy ARP* table to add or edit an IP address range.




# Physical interface

A FortiGate has several physical interfaces that can connect to Ethernet or optical cables. Depending on the FortiGate model, it can have a varying combination of Ethernet, small form-factor pluggable (SFP), and enhanced small form-factor pluggable (SFP+) interfaces.

The port names, as labeled on the FortiGate, appear in the interfaces list on the *Network > Interfaces* page. Hover the cursor over a port to view information, such as the name and the IP address.

Refer to Configuring an interface for basic GUI and CLI configuration steps.

For information about SFP/SFP+ interfaces, see:

- Transceiver status information for SFP and SFP+ interfaces on page 257
- GPON support for FortiGates with SFP/SFP+ ports NEW on page 258

## Transceiver status information for SFP and SFP+ interfaces

Transceiver status information for SFP and SFP+ interfaces installed on the FortiGate can be displayed in the GUI and CLI. For example, the type, vendor name, part number, serial number, and port name. The CLI output includes additional information that can be useful for diagnosing transmission problems, such as the temperature, voltage, and optical transmission power.

**To view transceiver status information in the GUI:**

1. Go to *Network > Interfaces*. The *Transceiver* column is visible in the table, which displays the transceiver vendor name and part number.
2. Hover the cursor over a transceiver to view more information.

**To view transceiver status information in the CLI:**

```
# get system interface transceiver
Interface port9 - SFP/SFP+
  Vendor Name  :            FINISAR CORP.
  Part No.     :            FCLF-8521-3
  Serial No.   :            PMS***
Interface port10 - Transceiver is not detected.
Interface port11 - SFP/SFP+
  Vendor Name  :            QNC
  Part No.     :            LCP-1250RJ3SRQN
  Serial No.   :            QNDT****
Interface port12 - SFP/SFP+
  Vendor Name  :            QNC
  Part No.     :            LCP-1250RJ3SRQN
  Serial No.   :            QNDT****
Interface s1 - SFP/SFP+
  Vendor Name  :            JDSU
  Part No.     :            PLRXPLSCS4322N
  Serial No.   :            CB26U****
Interface s2 - SFP/SFP+
  Vendor Name  :            JDSU
  Part No.     :            PLRXPLSCS4321N
  Serial No.   :            C825U****
Interface vw1 - Transceiver is not detected.
Interface vw2 - Transceiver is not detected.
Interface x1 - SFP/SFP+
  Vendor Name  :            Fortinet
  Part No.     :            LCP-10GRJ3SRFN
  Serial No.   :            19090910****
Interface x2 - Transceiver is not detected.
                                       Optical      Optical      Optical
SFP/SFP+     Temperature  Voltage      Tx Bias      Tx Power     Rx Power
Interface    (Celsius)    (Volts)      (mA)         (dBm)        (dBm)
------------ ------------ ------------ ------------ ------------ ------------
port9         N/A          N/A          N/A          N/A          N/A
port11        N/A          N/A          N/A          N/A          N/A
port12        N/A          N/A          N/A          N/A          N/A
s1            38.3         3.35         6.80         -2.3         -3.2
s2            42.1         3.34         7.21         -2.3         -3.0
x1            N/A          N/A          N/A          N/A          N/A
  ++ : high alarm, + : high warning, - : low warning, -- : low alarm, ? : suspect.
```

## GPON support for FortiGates with SFP/SFP+ ports NEW

FortiGates with SFP/SFP+ ports support Gigabit Passive Optical Network (GPON) transceivers, enabling the firewall to operate directly as an ONU (Optical Network Unit) and allowing deployment in fiber-based access networks. This allows customers transitioning from legacy DSL infrastructure to adopt widely used GPON technology, improving interoperability and easing migration.

Use the following diagnose commands:

| Command | Description |
|---|---|
| `diagnose gpon config loid [interface] [loid] [password]` | Configure Logical Identifier (LOID) username and password. |
| `diagnose gpon sysinfo [interface]` | Show transceiver system information. |
| `diagnose gpon state [interface]` | Show SFP GPON connection state. |
| `diagnose gpon traffic-stats [interface]` | Show SFP GPON traffic statistics. |
| `diagnose gpon reboot [interface]` | Reboot transceiver. |

**To detect a GPON transceiver:**

```
# get system interface transceiver sfp1
Interface sfp1 - SFP/SFP+/SFP28, 1000BASE-LX
  Diagnostics  :            temperature, voltage, Tx bias, Rx power (avg), Tx power
  Vendor Name  :            FORTINET
  Part No.     :            WST-SFP-GPAU2MIF
  Serial No.   :            TK5ANK0006
  Measurement  Unit         Value        High Alarm   High Warning Low Warning  Low Alarm
  ------------ ------------ ------------ ------------ ------------ ------------ ------------

  Temperature  (Celsius)     45.8         88.0         85.0        -40.0        -45.0
  Voltage      (Volts)       3.34         3.60         3.47         3.00         2.90
  Tx Bias      (mA)         14.49        60.00        55.00         0.00         0.00
  Rx Power Avg (dBm)         -9.2         -7.0         -8.0        -28.5        -29.6
  Tx Power     (dBm)          1.3          6.5          5.5         -0.5         -1.0
  ++ : high alarm, + : high warning, - : low warning, -- : low alarm, ? : suspect.
```

**To check the GPON system information:**

```
# diagnose gpon sysinfo sfp1
Mac Address: 80:A5:79:50:11:1D
Model Name: WST-SFP-GPAU2MIF
Vendor: FORTINET
Hardware Version: V1.0
Software Version: V1.2.4
Gpon SN: TK5ANK0006
```

**To check the GPON state:**

```
# diagnose gpon state sfp1
State codes: O1 - Initial, O2 - Standby, O3 - Serial-Number, O4 - Ranging
             O5 - Operation, O6 - POPUP, O7 - Emergency-Stop

los = 0
state = O5
fec = 1
```

**To check GPON traffic statistics:**

```
# diagnose gpon traffic-stats sfp1
dsDropEvents = 0
usDropEvents = 0
dsOcts = 14395972
usOcts = 34376
dsPkts = 224922
usPkts = 669
dsBcastPkts = 0
usBcastPkts = 0
dsMcastPkts = 0
usMcastPkts = 0
dsCrcErrPkts = 0
usCrcErrPkts = 0
dsUnderSzPkts = 0
usUnderSzPkts = 0
dsOverSzPkts = 0
usOverSzPkts = 0
dsFragments = 0
usFragments = 0
dsJabbers = 0
usJabbers = 0
dsPkts_l64 = 0
dsPkts_64_127 = 224920
dsPkts_128_255 = 0
dsPkts_256_511 = 2
dsPkts_512_1023 = 0
dsPkts_1024_1518 = 0
usPkts_l64 = 0
usPkts_64_127 = 0
usPkts_128_255 = 0
usPkts_256_511 = 2
usPkts_512_1023 = 0
usPkts_1024_1518 = 0
dsDiscards = 0
usDiscards = 0
dsErr = 0
usErr = 0
```

# VLAN

Virtual local area networks (VLANs) multiply the capabilities of your FortiGate and can also provide added network security. VLANs use ID tags to logically separate devices on a network into smaller broadcast domains. These smaller domains forward packets only to devices that are part of that VLAN domain. This reduces traffic and increases network security.

# VLANs in NAT mode

In NAT mode, the FortiGate unit functions as a layer-3 device. In this mode, the FortiGate unit controls the flow of packets between VLANs and can also remove VLAN tags from incoming VLAN packets. The FortiGate unit can also forward untagged packets to other networks such as the Internet.

In NAT mode, the FortiGate unit supports VLAN trunk links with IEEE 802.1Q-compliant switches or routers. The trunk link transports VLAN-tagged packets between physical subnets or networks. When you add VLAN subinterfaces to the FortiGate's physical interfaces, the VLANs have IDs that match the VLAN IDs of packets on the trunk link. The FortiGate unit directs packets with VLAN IDs to subinterfaces with matching IDs.

You can define VLAN subinterfaces on all FortiGate physical interfaces. However, if multiple virtual domains are configured on the FortiGate unit, you only have access to the physical interfaces on your virtual domain. The FortiGate unit can tag packets leaving on a VLAN subinterface. It can also remove VLAN tags from incoming packets and add a different VLAN tag to outgoing packets.

Normally in VLAN configurations, the FortiGate unit's internal interface is connected to a VLAN trunk, and the external interface connects to an Internet router that is not configured for VLANs. In this configuration, the FortiGate unit can apply different policies for traffic on each VLAN interface connected to the internal interface, which results in less network traffic and better security.

## Sample topology

In this example, two different internal VLAN networks share one interface on the FortiGate unit and share the connection to the Internet. This example shows that two networks can have separate traffic streams while sharing a single interface. This configuration can apply to two departments in a single company or to different companies.

There are two different internal network VLANs in this example. VLAN_100 is on the 10.1.1.0/255.255.255.0 subnet, and VLAN_200 is on the 10.1.2.0/255.255.255.0 subnet. These VLANs are connected to the VLAN switch.

The FortiGate internal interface connects to the VLAN switch through an 802.1Q trunk. The internal interface has an IP address of 192.168.110.126 and is configured with two VLAN subinterfaces (VLAN_100 and VLAN_200). The external interface has an IP address of 172.16.21.2 and connects to the Internet. The external interface has no VLAN subinterfaces.

When the VLAN switch receives packets from VLAN_100 and VLAN_200, it applies VLAN ID tags and forwards the packets of each VLAN both to local ports and to the FortiGate unit across the trunk link. The FortiGate unit has policies that allow traffic to flow between the VLANs, and from the VLANs to the external network.